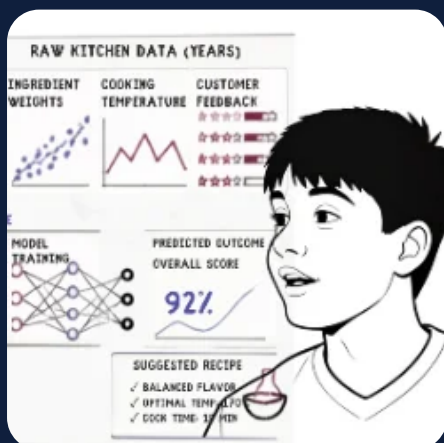


WHAT DOES AN

# ML Engineer do?

Here is a simple script (without corporate jargon) — read the story aloud at bedtime, or use it to spark a fun conversation right at the dinner table. We've also added real world examples and two activities to make the concept interesting and memorable.



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# Imagine a kitchen that invents its own recipes.

Most cooks just follow a cookbook. But this kitchen has someone who invents brand new recipes by studying years of raw kitchen data — millions of old receipts, exact ingredient weights, cooking temperatures, and customer feedback.

**The person who does this is the ML Engineer.**

You can think of them as the Kitchen Food Scientist and Recipe Inventor.

They don't just buy a pre-made cookbook. Instead, they spend hours cleaning up messy old data, then feed it into complex math to train the kitchen's brain from scratch — until it can invent recipes on its own.



**Then, they test their theory.**

They run experiment after experiment, checking whether the kitchen's guesses turn out right, and keep adjusting the math until the recipe reliably tastes good.

**At a company like Roblox, the ML Engineer builds the hidden math engines behind the game. They study millions of hours of gameplay to learn which obstacle-course jumps are too hard, too easy, or just right for kids.**

**ML engineers are like kitchen food scientists — inventing recipes from data instead of just following them.**

# ML Engineer *[noun]*

The people who teach computers to find patterns in huge piles of information, then use what the computer learned to make smart predictions. They don't write down rules by hand — they let the data train the model.

IN THE STORY

**Kitchen Food  
Scientist & Recipe  
Inventor**

=

IN THE REAL WORLD

**ML Engineer**

Think of them as scientists running experiment after experiment on data until the recipe learns to cook itself.

WHAT AN ML ENGINEER ACTUALLY DOES

Gather data

→

Clean it

→

Train model

→

Test it

→

Predict

# Pick the ones your child knows.

Part 1 of 2



## Roblox

ML Engineers study millions of hours of gameplay to learn which obstacle-course jumps are too hard, too easy, or just right for kids.



## Gmail Spam Filter

An ML Engineer trains a model on millions of old emails so it learns to spot spam and keep it out of your inbox.



## Card Fraud Alerts

Banks train models on years of past transactions so they can flag a strange purchase the moment it happens.

# Pick the ones your child knows.

Part 2 of 2



## Weather Forecasts

An ML Engineer trains a model on years of past weather data so it can predict tomorrow's forecast.



## Face Unlock

An ML Engineer trains a model on patterns in faces so your phone recognizes you instantly and unlocks.



## Amazon Recommendations

An ML Engineer trains a model on millions of past purchases so it can guess what you might want to buy next.

# Match the following

Draw a line from each thing in the kitchen to the identical concept at the tech company.

## IN THE KITCHEN

## AT THE TECH COMPANY

1

**The Box of Old Receipts**

Years of records the scientist studies to find patterns.

2

**The Taste-Test Panel**

Checks whether a new recipe actually tastes right.

3

**The Exact Measuring Scale**

Precise weights, so the numbers can be trusted.

4

**The Food Scientist**

Trains the kitchen's brain from raw data.

A

**ML Engineer**

Trains the model from raw data.

B

**Training Data**

Past records the model studies to learn patterns.

C

**Data Accuracy**

Precise, error-free numbers the model can rely on.

D

**Model Testing**

Checks whether the model's predictions are actually right.

# Quiz time

There are no wrong answers — just fun guessing!

## QUESTION 1 · THE BAD RECIPE BOOK

If a food scientist only studies burnt experiments, the recipes they invent will taste bad. What do you think happens if an ML Engineer trains a model on bad or unfair data?

## QUESTION 2 · THE TASTE-TEST PANEL

A food scientist always taste-tests a new recipe before serving it to guests. Why do you think an ML Engineer always tests a model before it goes out to real users?

## QUESTION 3 · THE BRAND NEW SNACK

If a food scientist wants to invent a brand new snack nobody has tried, what kind of data would they need to study first?

## QUESTION 4 · BUILD YOUR OWN LAB

If you were an ML Engineer building a brand new model, what is the #1 thing you would want it to learn to predict?

# Answers

## ANSWER 1

The model will learn the wrong lessons and make bad predictions — just like a recipe built only on burnt experiments will taste bad. That is why ML Engineers work hard to gather clean, fair data.

## ANSWER 2

Without testing, a model might make silly or unsafe guesses once it meets real people. Testing catches mistakes early, just like a taste test catches a bad recipe before it reaches the guests.

## ANSWER 3

They would need data on flavors people already enjoy, past experiments that worked and failed, and feedback from taste testers — the same way an ML Engineer studies existing data before inventing something new.

## ANSWER 4

Let them get creative! Examples: predicting which game level a player will enjoy, spotting a fake review, or guessing tomorrow's weather correctly.

## FOR THE MATCHING GAME

**1 – B   2 – D   3 – C   4 – A**